

Logan Kay

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Experience

Firmware Engineering Intern – Allegion – Indianapolis, Indiana **May 2025 – Aug 2025**

- Developed a C based, event driven control algorithm within a FreeRTOS environment for a wireless application running on Silicon Labs MGM240S microcontrollers, integrating multiple component drivers (SPI and I2C) and hardware peripherals
- Utilized industry standard tools including Git for version control, Simplicity Studio for development and debugging, and FreeRTOS for multitasking, peripheral coordination, and inter-task communication
- Incorporated low-power techniques through MCU sleep mode management as well as real time feedback mechanisms via several GPIO driven peripherals and a UART based debug terminal for runtime information and interactive testing
- Collaborated with external vendors to integrate third-party modules and ensure hardware/software compatibility, matching technical requirements and resolving interfacing issues

Hardware Engineering Intern – Allegion – Indianapolis, Indiana **May 2024 – Aug 2024**

- Developed detailed circuit schematics for electronic devices, ensuring compliance with industry and company standards
- Created high precision PCB layouts for electronic projects, implementing advanced techniques to optimize space, enhance functionality, and maximize efficiency.
- Designed PCB for programming board to take USB-C power input and enable device debugging by receiving data over BLE and communication of that data through UART
- Assembled and soldered the programming board project from the ground up using reflow and hand soldering techniques, including surface-mount and through-hole components

Projects

CLI for Pre-Trained Model Analysis – ECE 30861/461 – Purdue University **Nov 2025**

- Developed a Python-based CLI tool within a small SWE team to evaluate pre-trained AI/ML models from Hugging Face and GitHub across metrics like license compliance, dataset/code availability, performance claims, etc.
- Designed a modular, test-driven architecture (pytest) with a structured NDJSON output, and parallelized metric computation with per metric and overall latency tracking across ingestion of multiple model URLs
- Designed and deployed a frontend/backend REST API via AWS EC2 and ECS with fully automated testing via CI/CD to enable CRUD capabilities for pre-trained AI/ML model interaction

User Authentication Web Application – ECE 30864 – Purdue University **Feb 2025**

- Built a full-stack Flask web app using SQLite integration to manage user accounts and records as well as user-generated posts
- Designed and implemented relational database schemas and SQL queries for multiple database synchronization
- Integrated dynamic routing and Jinja2 templating to enable user-specific profile along with content creation and deletion functionality as well as hashing via SHA-256 for secure user identification

Multithreaded Polynomial Calculator– ECE 39595 – Purdue University **Nov 2024**

- Designed and implemented a C++ application to perform polynomial computations, leveraging multithreading to parallelize processing and improve performance
- Structured thread execution to avoid race conditions by using thread-local storage, ensuring data integrity
- Integrated modularity and object-oriented design by encapsulating polynomial operations in classes, improving code reusability

Precision Pulse STM32 Driven Game– ECE 362 – Purdue University **Nov 2024**

- Led the configuration of the SPI interface and integration of an OLED display to support menu navigation and real-time visualization of game state and player score
- Led the design and implementation of the interrupt-driven game control logic using GPIO-based EXTI for multiple button inputs, and integrated ADC input from a potentiometer to allow users to dynamically set game difficulty
- Assisted in the integration of additional peripheral components such as I2C for EEPROM read/write functionality and high score tracking as well as GPIO, DMA, and hardware timers to drive an RGB LED matrix for the “moving wave” output

Skills

Languages: C, C++, Python, JavaScript, HTML, SystemVerilog, MATLAB, SQLite, Bash, Assembly

Software: AWS(DynamoDB, EC2, ECS), Git/Github, FreeRTOS, Jira, Altium Designer, Simplicity Studio, Flask

Hardware: ESP32, STM32, MGM240x, JTAG/SWD, I²C, SPI, UART, GPIO, ADC, DAC, PWM

Activities

Purdue School of ECE – ECE 570(Artificial Intelligence) Undergraduate TA **Aug 2025 – Dec 2025**

Purdue Club Tennis – Travel Team Member **Aug 2023 – Current**

Sigma Pi Fraternity – Technology Chair, Rush Chair, Intramurals/Workout Chair – Purdue University **Aug 2022 – Current**

Education

Purdue University (Honors) – Bachelor of Science in Computer Engineering **Aug 2022 – May 2026**

Minor: Economics | **Concentrations:** SWE, AI/ML

GPA: 3.66/4.0